1. **Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.**

Step-by-step procedure

1. Connect to Wi-Fi: Connect your computer to your home Wi-Fi network.
2. Open Command Prompt: Press Windows + R, type cmd, and press Enter.
3. Find the Router IP: Type the following command and press Enter
4. Check Default Gateway: Find the Default Gateway under your connected Wi-Fi adapter. It may be 192.168.1.1 or 192.168.0.1.
5. Open a Browser: Open Chrome, Edge, or another browser.
6. Enter the Router IP: Type the Default Gateway address in the address bar, for example: http://192.168.1.1
7. Open the Login Page: Press Enter. The router's admin login page should appear.
8. Take a Screenshot: Press Windows + Shift + S, select the login page area, and save the screenshot for your assignment.



Result

- The router's administration login page was opened using its Default Gateway IP address, and a screenshot was captured for the assignment.

2. Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

- Method 1: Windows Laptop
  - Connect your laptop to your Wi-Fi network.
  - Press Windows + R.
  - Type cmd and press Enter.
  - Type the following command:
    - ipconfig
  - Find your connected Wireless LAN adapter Wi-Fi section.
  - Note the IPv4 address, subnet mask, and default gateway.
- Observation Table

NETWORK SETTING	VALUE
IP ADDRESS (IPV4)	_____
SUBNET MASK	_____
DEFAULT GATEWAY	_____

Example values (your actual values may differ):

NETWORK SETTING	EXAMPLE
IP ADDRESS	192.168.1.105
SUBNET MASK	255.255.255.0
DEFAULT GATEWAY	192.168.1.1

3. Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access. Remember to revert to automatic/DHCP after testing to avoid network issues.

1. Open Settings: Press Windows + I to open Windows Settings.
2. Open Network Settings: Select Network & internet.
3. Select Connection: Click Wi-Fi if using Wi-Fi, or Ethernet if using a cable.
4. Open IP Settings: Click your connected network and find IP assignment.
5. Edit IP Assignment: Click Edit and change Automatic (DHCP) to Manual.
6. Enable IPv4: Turn on the IPv4 option.
7. Enter Static IP Details: If your router uses 192.168.1.1 and your network uses the 192.168.1.x range, enter:
    - IP address: 192.168.1.50
    - Subnet mask: 255.255.255.0
    - Gateway: 192.168.1.1
    - Preferred DNS: 8.8.8.8
    - Alternate DNS: 8.8.4.4

    Important: Use these example values only if they match your network. Check your router's actual gateway and subnet first, and ensure the chosen IP is unused and outside the DHCP pool or reserved for your device.
8. Save Settings: Click Save to apply the static IP configuration.
9. Test Internet Access: Open a web browser and visit a website. You can also open Command Prompt and run:
   ping 8.8.8.8
10. Record the Result: Note whether the internet works with the static IP.
11. Revert to DHCP: Return to IP assignment → Edit, select Automatic (DHCP), and click Save.

➢ **Observation table**

Test	Result
Internet using DHCP	_____
Internet using static IP	_____
IP configuration restored to DHCP	Yes / No

4. In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

1. Connect your laptop to your home Wi-Fi network.
2. Open Command Prompt and type `ipconfig`.
3. Note the Default Gateway address.
4. Open a web browser and enter the Default Gateway IP address.
5. Log in to the router's admin panel.
6. Navigate to DHCP Settings or LAN Settings.
7. Find the DHCP Start IP Address and DHCP End IP Address.
8. Write down the current range shown on your router.

Observation

Setting	Value (example only)
DHCP Start IP	`192.168.1.100`
DHCP End IP	`192.168.1.200`
DHCP Range	`192.168.1.100 – 192.168.1.200`

Replace these example values with the actual values displayed in your router.

What happens if two devices have the same IP address?

If two devices are assigned the same IP address on the same local network, an IP address conflict occurs. This can cause intermittent connectivity, failed communication, or loss of network access for one or both devices.

Result

The DHCP range was identified in the router's administration panel, and the effect of duplicate IP addresses was understood.